| | VECTOR SHORTLISTING TEST – MAY-2024 | | |
|---|---|---|---|
| VECTOR INSTITUTE | VECTOR ID: | DATE : 25-04-2024 | MARKS |
| | NAME: | TIME : 3 hrs | |
| | | MARKS : 25 M | |

## Module: C

1) [1M]

```
#include<stdio.h>
#define VAL 32
int main()
{
    char arr[]="vectorindia";
    *(arr+0)&=~VAL;
    *(arr+6)&=~VAL;
    printf("%s\n",arr);
}
```

2) [1M]

```
#include<stdio.h>
int main()
{
    signed int x;
    x=(64 >> (2+1-2)) & (~(1<<2));
    printf("%d\n",x);
}
```

A) 16 B) 64 C) 32 D) -64

3) [1M]

```
#include<stdio.h>
int main()
{
    int x = 48;
    int y = 0x48;
    int z = 048;
    printf("%d %d %d\n",x,y,z);
```

```
}
```

4) [1M]

```
#include<stdio.h>
int main()
{
    char *ptr = "Compassion";
    char arr[] = "Compassion";

    if(ptr == arr)
    printf("Equal\n");
    else
    printf("Not Equal\n");
}
```

5) [1M]

```
#include<stdio.h>
int main()
{
    int x = 300;
    char *p = (char *)&x;
    int **ptr = (int **)&p;

    printf("x = %d  *p = %d  **ptr = %d\n",x,*p,**ptr);
}
```

6) [2M]

```
#include<stdio.h>
#include<string.h>
int main()
{
    char str[] = "abcdef";
    char *ptr = str+strlen(str)-1;

    printf("%c\n",*ptr--);
    printf("%c\n",*--ptr);
    printf("%c\n",(*ptr)--);
    printf("%c\n",--*ptr);
    printf("%c\n",--*--ptr);
    printf("%s\n",str);
}
```

7) [1M]

```
#include<stdio.h>
int main()
```

```
{
    char *p[] = {"Promise","Tolerance","Character","Responsibility"};
    char **ptr = p;
    ptr += 2;
    printf("%s\n",(*ptr)+4);
}
```

**8)** [1M]

```
#include<stdio.h>
int main(){
int a=0;
{
        int a=10;
        printf("%d\n",a);
        a++;
        {
                a=20;
        }
        {
                printf(" %d\n",a);
                int a=30;
                {
                        a++;
                }
                printf(" %d\n",a++);
        }
        printf(" %d\n",a++);
    }
        printf(" %d\n",a);
        return 0;
}
```

**9)** [1M]

```
#include <stdio.h>
void fun(int n)
{
    if(n > 0)
    {
        fun(--n);
        printf("%d ", n);
        fun(n-1);

    }
```

```
}
int main()
{
    fun(3);
    return 0;
}
```

10) [1M]

```
#include<stdio.h>
int main()
{
        char str[] = "abcdefgh";
        int *p = (int *)str;
        printf("%x  %x\n",p[0],p[1]);
}
```

11) Define the following things. [3M]
   i) Array of pointer
   ii) Pointer to an array
   iii) Function pointer

12) What are the difference between structures and unions. [2M]

13) Write a program to count the number of words in a given file. [4M]

14) Write only a function to reverse the data in the single linked list. [5M]

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## Module: C++

1) _____ means that both the data and the methods which may access it are defined together in the same unit. (1m)

a) Data hiding b) Encapsulation c) Data Binding d) None

2) Can I create new operators using operator overloading? (1m)

a) Yes b) No

3) Can I inherit the constructor and destructor of a base class? (1m)

a) Yes b) No

4) The _____ of a function is not a factor in distinguishing overloaded functions. (1m)

5) (2 m)

```
int sum(int a, int b=5, int c=10);
main()
{
    cout<<sum(5)<<sum(10,10)<<sum(10,10,20);

}
int sum(int a, int b, int c)
{
return a+b+c;
}
```

What is the output of the above code?

6) List the different types of constructors in C++? (2m)

7) Pick the correct statement about references. (1m)

a) References can be assigned value NULL
b) References once assigned cannot be changed to refer another variable
c) Reference should not be initialized when created
d) Reference is the same as pointers

8) How constructors are different from other member functions of the class? (1m)

a) Constructor has the same name as the class itself
b) Constructors do not return anything
c) Constructors are automatically called when an object is created
d) All of the mentioned

9) (1m)

```
#include<iostream>
using namespace std;
namespace ABC
{
int data=10;
void foo()
{
cout<<"in ABC foo :"<<endl;
}
}
namespace XYZ
{
float data=2.3;
void foo()
{
cout<<"in XYZ foo:"<<endl;
}
}
using namespace XYZ;
int main()
}
cout<<data<<endl;
ABC::foo();

}
```

(1m)

10) The mechanism of deriving a class from another derived class is know as _____ (1m)

a) multiple inheritance
b) multilevel inheritance
c) single inheritance
d) hierarchical inheritance

(1m)

11) Syntax for Pure Virtual Function is _____

12) The redefinition of a virtual function by a derived class is called_____ (1m)

a) overriding
b) overloading
c) template
d) generic function

13) which of the following concept is used to implement late binding? (1m)

a) virtual function
b) operator function
c) const function
d) static function

14) Difference between Function overloading and Function templates (2m)

15) If abstract class is inherited by derived class, then_____ (1m)

a) Derived class should provide definition for all the pure virtual functions
b) Derived class also become abstract if fails to implement pure virtual functions
c) Objects of derived class can't be created if it fails to implement pure virtual functions
d) All of these

16) Write the syntax for post increment operator overloading using friend function (1m)

17) _____ class makes the single members copy from base class to final derived class in multipath inheritance to avoids duplication copy in final derived class.

(1m)

18) write a program to design stack operations using class Template. (5M)

**NOTE**:store and retrieve any type of datatype.

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## Module: ARM

1) Explain about Volatile Keyword (2M)

2) The maximum memory interfaced or connected to Microcontroller depends on size of the _____ bus. (1M)

3) How many power control pins are there in LCD ? (1M)

a) 2 b ) 1 c ) 3 d) 4

4) ARM7 is following _____ mapped I/O system. (1M)

5) In _____ & _____ pins for lpc2129 has TXD0 & RXD0.

a) P0.0 & P0.1 b) P0.1 & P0.0 c) P0.8 & P0.9 d) P0.9 & P0.8

6) The 32 bit PC is a counter which is incremented to the value stored in _____. (1M)

7) _____ is the LCD command for Display ON, Cursor Blinking . (1M)

8) Standard speed In I2C protocol is_____ . (1M)

a) 1 Mbps b) 5 Mbps c) 3.4 Mbps d) 100Kbps

9) To get 3 seconds delay from timer0, (i.e T0PR value should be) T0PR= _____. Assume that PCLK is 15MHz. (1M)

10) LPC2129 has _____ timers. (1M)

a) two 32-bit timers b)two 16-bit timers c)one 32-bit timer d)one 16-bit timer

11) Write the code snippet to Allocate Slot1 to External Interrupt 1 ? (1M)

12) The ARM7 is the type of _____ & _____ architecture. (1M)

13) MCP3204 supports _____ & _____ modes in SPI. (1M)

14) Explain about SOF(Start Of Frame) bit in CAN ? (1M)

15) What is the use of ADC ?

16 ) Explain about I2C start condition ? (2M)

**B) Descriptive** **(7 Marks)**

17) Write an ECP to blink LED1 continously for every 500 ms & power-on LED0 when the SW switch is pressed & power-off LED0 when SW switch is released ? 5 M

18) Explain about CAN Error states ? 2 M

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## Module: LINUX

1.fork system call returns ---------- to child process and ---------- to parent process? (1m)

2.Signals that we can not catch or ignore are ? (1m)

3.kill(getpid(),signalno) is equivalent to raise() function?(T/F) (1m)

4.what is the default action of SIGCHLD signal---------- (1m)

5.----------calls are used to replace current process image with new process image and with same pid and ppid? (1m)

6.if umask value is 0057 then new files are created with permissions are? (1m)

7.soft link files inode numbers are same(T/F) (1m)

8.write the prototype of the following functions? (2m)
1:open() 2:read()

9.what is the difference between F_SETLKW and F_SETLK in fcntl function? (2m)

10.---------- function used to create a named pipe? (1m)

11.by closing the read end of pipe if any process writing data ---------- signal is generated? (1m)

12.what is the command used to display the message queue info in a system? (1m)

13.shared memory is asynchronous IPC?(T/F) (1m)

14.semop() system call third argument represents? (2m)

15.define critical section of code? (2m)

16.What is internal fragmentation? (2m)

17.write differences between process and threads? (3m)

18.----------function used to get the thread id with in the thread? (1m)

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## Module: TCPIP

**( A) Fill in the blanks**

1)_____ layer responsibility to encrypt and decrypt the user level data In OSI reference model?

2) _____is the maximum size of the TCP header?

3) IPv6 has ___ bit addresses.

4) The first octet range for a class A IP address is _____.

5) Header size of the UDP is _____

6) A Gate way works in _____ layer

7)_____protocol is used to send the mail?

8) Default netmask for class A is_____

9) The maximum size of a datagram of Ipv4 is_____.

10) size of port number is_____.

11) The loop-back address is used to send a packet from the ____ to ___ on the network.

**(B) MCQ's ( 8 marks)**

1) The physical layer is concerned with the transmission of _____over the physical medium.
a) programs b) dialogs c) protocols d) bits

2) What is the PDU In internet layer?

a) IP datagram b) UDP datagram c) segment d) none

3) The PDU of physical layer in TCP/IP

a) data b) bit c) frame d) packet

4) In class B IP address how many bits are allocated for network ID?

a)4bit b) 8bit c) 32bit d) 16bit

5) MAC addresses are also known as?

a) Hardware address b) physical address c) Hardware & physical address d) IP address

6) what is the range of class D IP?

a) 224.0.0.0 – 239.255.255.255 b) 128.0.0.0 – 191.255.255.255
c) 0 – 127.255.255.255 d) 192.0.0.0 – 223.255.255.255

7) PORT number used for HTTP

a) 80 b) 20 c) 21 d) 112

8) Identify the class of following IP address: 4.5.6.7

a) class A b) class B c) class C d) class D

**(C) Each question 2 Marks**

1)what is socket ? Types of socket.
2) Write the prototype for a) bind() b) listen()
3) Differentiate between MAC and IP address.